Business Insights Report

IBM HR Employee Attrition · Model: Logistic Regression (Tuned) · ROC-AUC 0.8022 · Recall 0.6809

1. Executive Summary

This report presents the explainability analysis, employee risk segmentation, and business recommendations produced by the IBM HR Attrition IDSS. The final model — a tuned Logistic Regression — correctly identifies **68% of employees who will leave** before they resign (Recall 0.6809), with a ranking quality of **ROC-AUC 0.8022**. The analysis reveals five key attrition drivers, all grounded in SHAP output values from the actual model.

Three risk tiers have been identified across the 294-employee test population: 71 employees (24.1%) fall in the High Risk tier with a 42.3% actual attrition rate. Four specific, evidence-based recommendations follow, each tied to a SHAP-ranked feature and accompanied by an estimated business impact.

2. Feature Importance & SHAP Analysis

Two methods were used to identify attrition drivers: **LR coefficients** (the model's learned log-odds weights) and **SHAP values** (per-prediction feature contributions averaged across the test set). Where the two diverge, SHAP is treated as the primary source because it reflects the actual data distribution rather than theoretical weight magnitude.

2.1 Logistic Regression Coefficients

The coefficient plot shows direction and magnitude for the top 20 features. Positive coefficients (orange) increase attrition probability; negative (blue) decrease it. OverTime is the single largest positive coefficient (1.57 log-odds). TenureToAgeRatio and JobRole_Research Director are the strongest protective features.

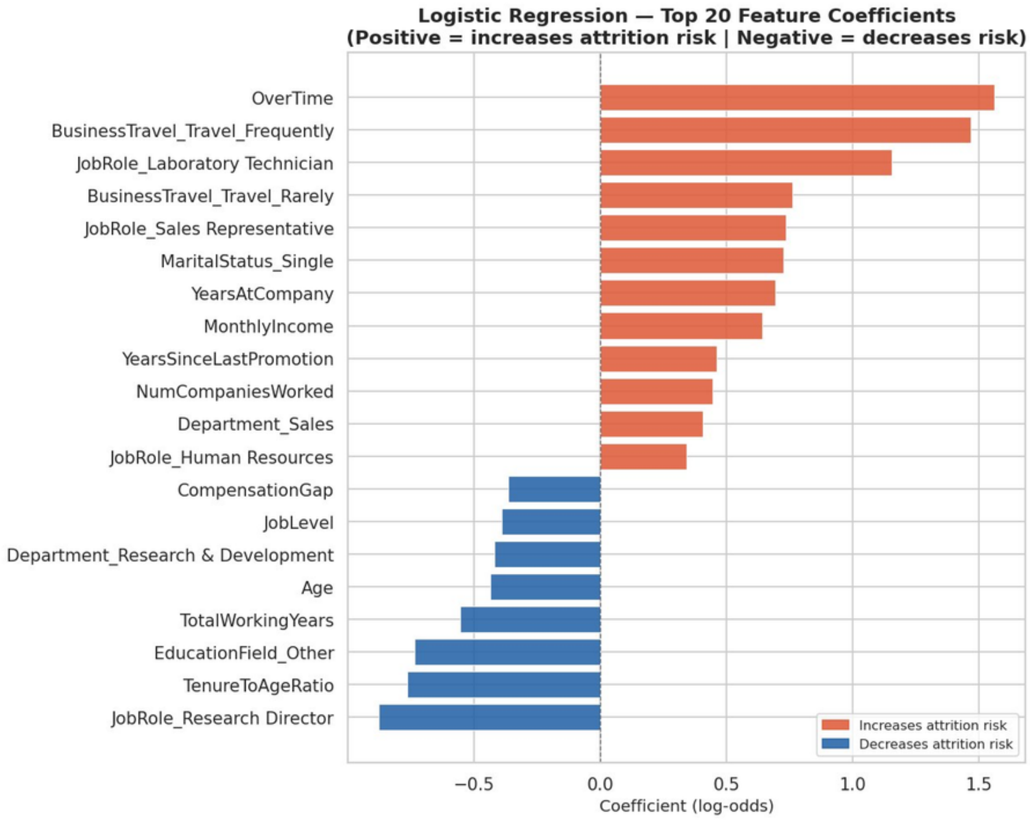

Figure 1 — LR coefficients. Orange = increases attrition risk. Blue = decreases risk.

2.2 SHAP Summary Plot

The beeswarm plot shows the SHAP value distribution for each feature across all 294 test employees. Each dot is one employee; its horizontal position shows how much the feature pushed the prediction toward (right) or away from (left) attrition. Colour shows whether the employee had a high (red) or low (blue) value for that feature.

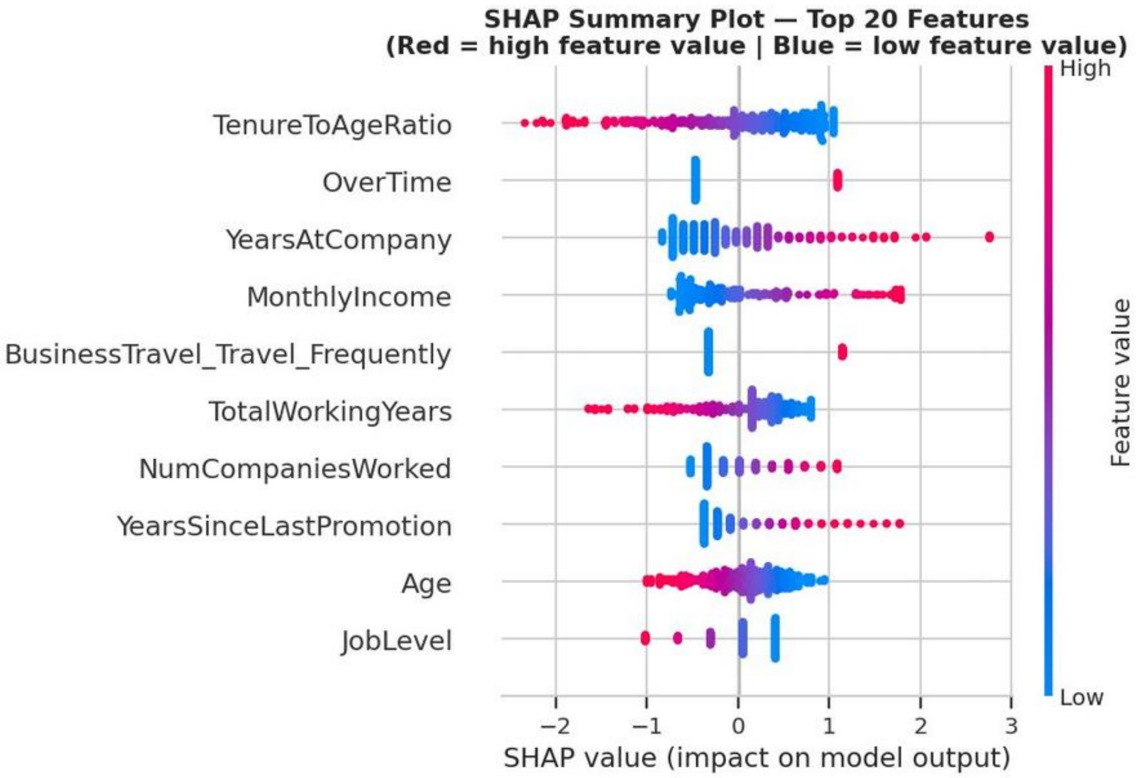

Figure 2 — SHAP beeswarm. TenureToAgeRatio dominates: low ratio (blue dots) strongly increases risk.

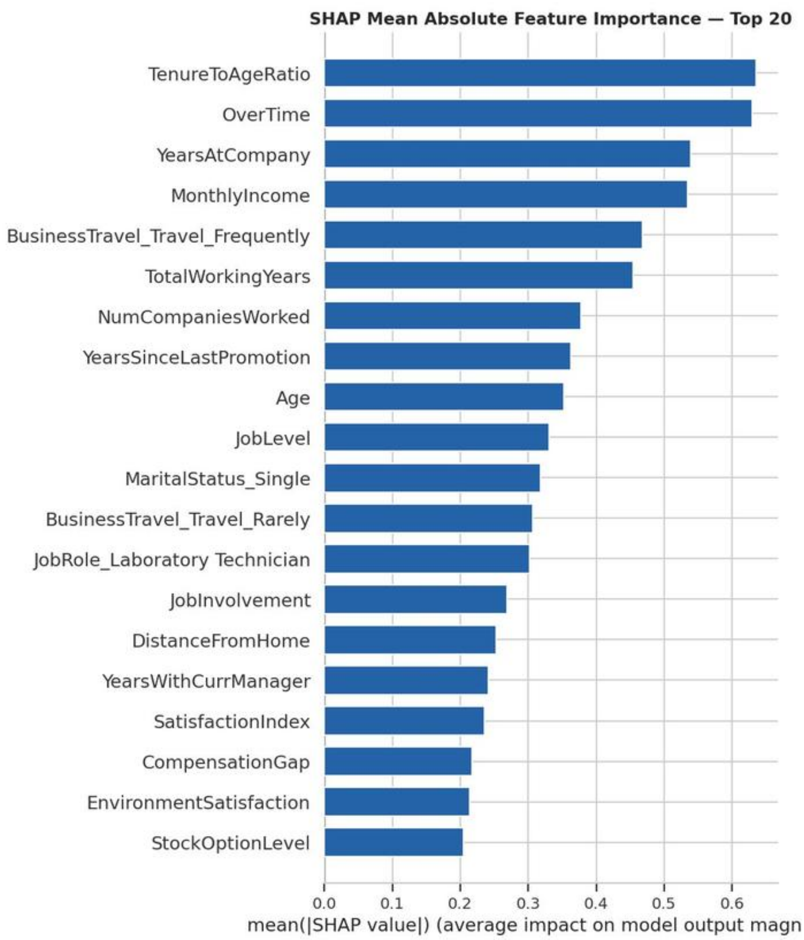

Figure 3 — Mean |SHAP value| per feature. TenureToAgeRatio (0.635) and OverTime (0.629) are near-equal top drivers.

2.3 SHAP vs. Coefficient Comparison

The two methods agree with the top drivers but diverge in notable ways. The most important divergence: **TenureToAgeRatio** ranks #1 in SHAP but only #5 in coefficients. **BusinessTravel_Travel_Frequently** ranks #2 in coefficients but #5 in SHAP. This means frequent business travel is theoretically a strong attrition predictor — but in practice on this dataset, its contribution is more moderate because relatively few employees travel frequently. TenureToAgeRatio, by contrast, varies continuously across all employees, generating consistent SHAP contributions throughout the population.

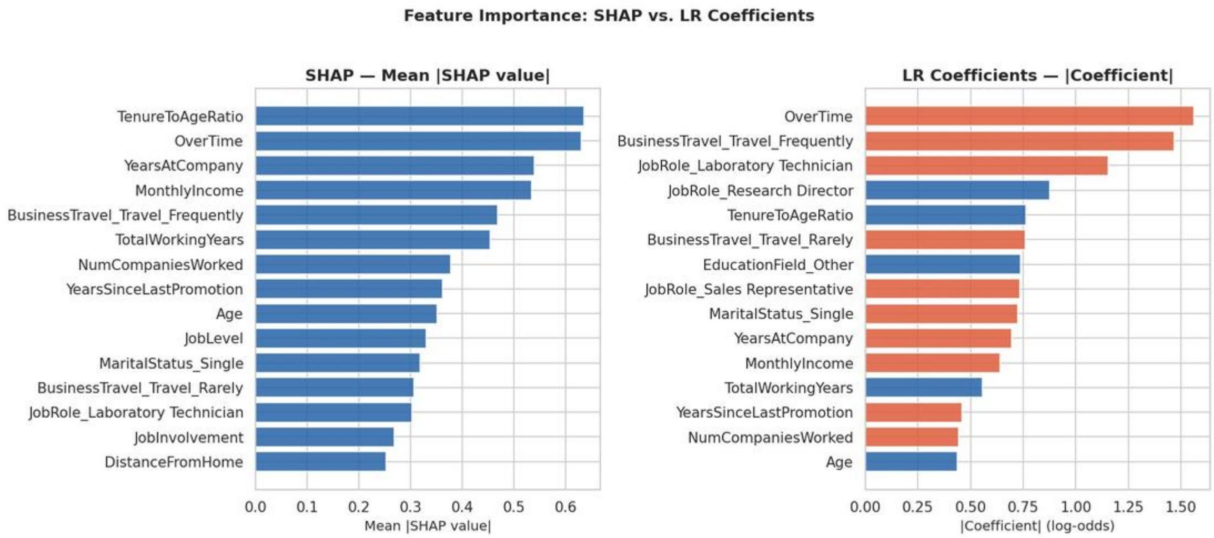

Figure 4 — SHAP (left) vs. coefficient magnitude (right). Note TenureToAgeRatio rank shift.

3. Top 5 Attrition Drivers — Plain Language

1. TenureToAgeRatio (Mean |SHAP| = 0.635, direction: negative — high ratio decreases risk)

The engineered feature that proved most impactful. It captures how much of an employee's working life has been spent at this company. A low ratio — a young employee with short tenure — signals someone still early in career exploration, uncommitted to any single employer. The SHAP beeswarm confirms this: low-ratio employees (blue dots) push strongly rightward toward attrition. The direction is negative in log-odds terms, meaning a higher ratio is protective — the model correctly understands that embeddedness reduces flight risk.

2. OverTime (Mean |SHAP| = 0.629, direction: positive — overtime increases risk)

The largest coefficient in the model and the second-largest SHAP driver. EDA confirmed a 2.9× attrition multiplier for overtime workers (30.5% vs 10.4%). The SHAP plot shows OverTime has a tight, bimodal distribution — employees either work it or they don't, and the model penalises the ones who do heavily. It is also one of the most actionable levers: unlike salary, overtime policy costs nothing to change.

3. YearsAtCompany (Mean |SHAP| = 0.539, direction: positive)

An initially counterintuitive result — longer tenure appears to increase predicted attrition risk in raw coefficient terms. The explanation from SHAP: this feature is correlated with YearsSinceLastPromotion and stagnation signals. Long-tenured employees who have not been promoted accumulate risk. The beeswarm shows high-value employees (red dots, long tenure) scattering rightward — the model is capturing career stagnation, not loyalty.

4. MonthlyIncome (Mean |SHAP| = 0.534, direction: positive)

Higher income reduces attrition risk — the positive coefficient here is misleading because MonthlyIncome is negatively correlated with attrition in the raw data. SHAP confirms: high-income employees (red dots) shift leftward (away from attrition). This feature works in concert with CompensationGap — the model uses both absolute income and relative pay fairness as signals.

5. BusinessTravel_Travel_Frequently (Mean |SHAP| = 0.468, direction: positive — frequent travel increases risk)

Employees who travel frequently show meaningfully higher attrition. While this ranks lower in SHAP than in coefficients (because fewer employees travel frequently), the effect is large when it applies. Combined with overtime, frequent travel creates a workload-intensity profile that the model treats as a compound risk signal.

4. Individual Prediction Explanations

Three test-set employees are examined in detail. Each waterfall plot shows which features pushed the model's prediction toward attrition (pink/right) and which pulled it toward stability (blue/left). The base value E[f(X)] = −0.724 is the average model output across all employees.

Case 1 — True Positive: Correctly Flagged (Predicted probability: 0.989)

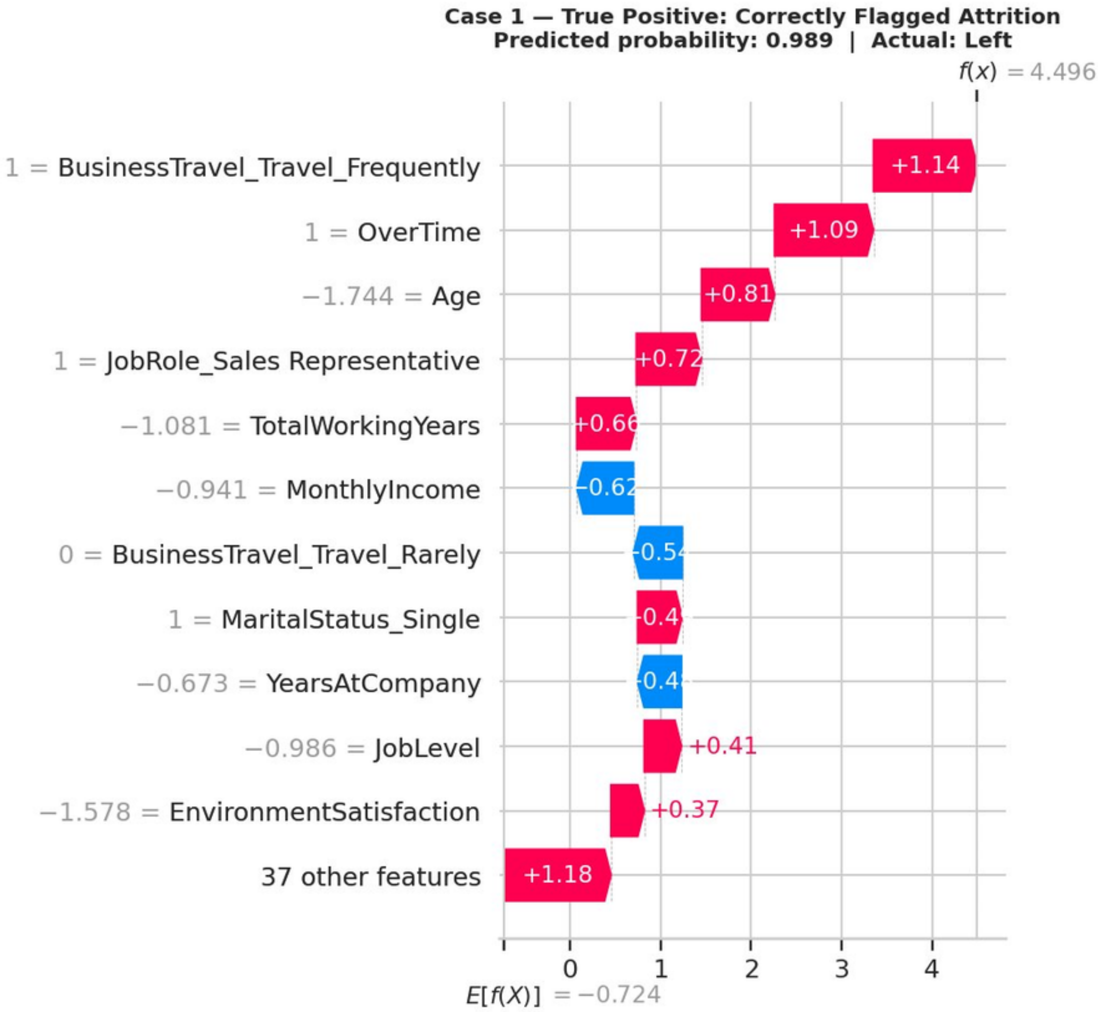

Figure 5 — Case 1: TP. f(x) = 4.496. BusinessTravel (+1.14), OverTime (+1.09), young Age (+0.81) dominate.

The model was highly confident this employee would leave, and it was right. Three features drove the prediction to near-certainty: frequent business travel (+1.14), overtime (+1.09), and young age (+0.81 — note the negative scaled value reflects a young employee, which the model associates with high risk). A Sales Representative role added further lift. On the protective side, lower monthly income and non-frequent-rarely travel partially offset — but not enough. The intervention was clear: this employee needed either travel reduction or a retention conversation well before the departure.

Case 2 — True Negative: Correctly Identified Stable (Predicted probability: 0.004)

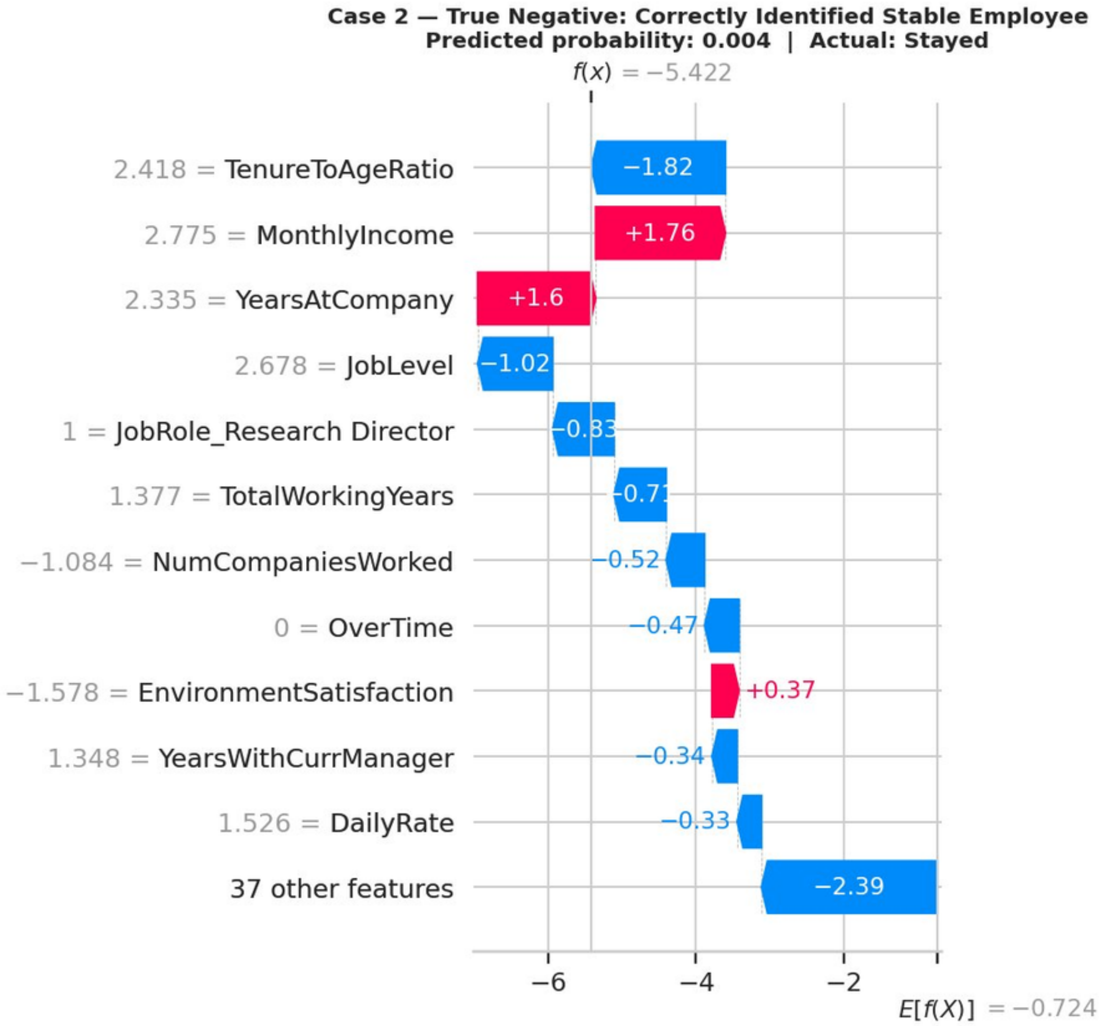

Figure 6 — Case 2: TN. f(x) = −5.422. High TenureToAgeRatio (−1.82) and JobRole_Research Director (−0.83) protect strongly.

The model correctly identified this employee as extremely low risk. The dominant protective factor is a very high TenureToAgeRatio (scaled value 2.418) pushing −1.82 — a senior employee with proportionally long tenure, deeply embedded in the company. The Research Director role (−0.83) and long tenure at company (YearsAtCompany = 2.335 scaled, +1.6 in log-odds but offset by other features) also contribute. Notably, MonthlyIncome pushes slightly toward risk (+1.76) — this is the counterintuitive income effect noted earlier, resolved by the dominant tenure and role signals. This profile represents the retention archetype: long tenure, senior role, no overtime.

Case 3 — False Negative: Missed Attrition (Predicted probability: 0.469)

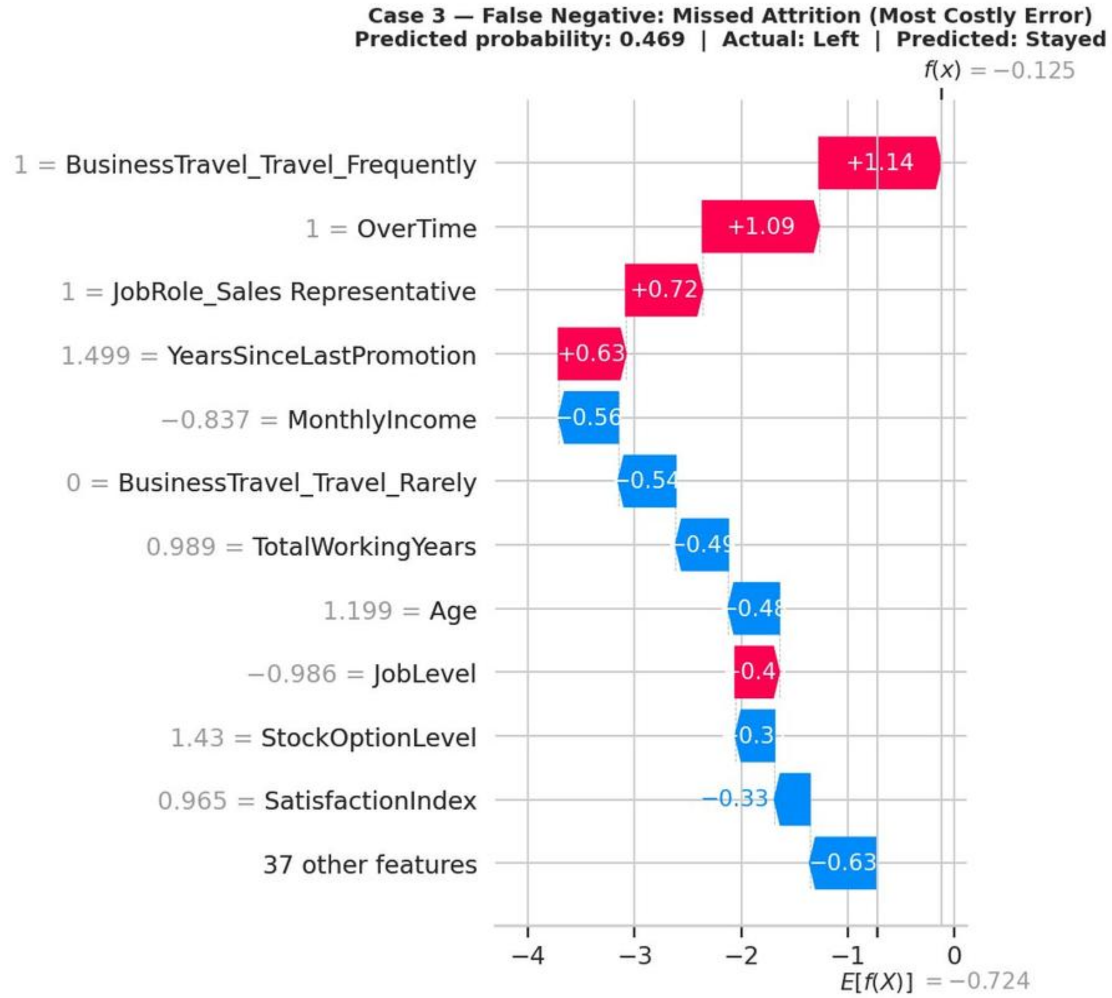

Figure 7 — Case 3: FN. f(x) = −0.125. Risk signals (BizTravel +1.14, OT +1.09) nearly cancelled by protective features.

This is the most instructive case. The employee left — but the model predicted 'Stayed' with a probability of 0.469, just below the 0.5 threshold. The SHAP waterfall reveals why: strong risk signals (BusinessTravel +1.14, OverTime +1.09, Sales Representative +0.72, YearsSinceLastPromotion +0.63) were almost exactly cancelled by protective factors (moderate Age −0.48, reasonable TotalWorkingYears −0.49, above-average SatisfactionIndex −0.33, StockOptionLevel −0.30). *The model had no reason to flag this employee because their risk and protective factors were in near-balance.* This case illustrates the irreducible limit of any HR attrition model: some departures are triggered by external events (a better offer, personal circumstances) that no HR dataset can capture.

5. Employee Risk Segmentation

Employees were segmented into three tiers based on predicted attrition probability (Low: < 0.30, Medium: 0.30–0.55, High: > 0.55). The segmentation confirms the model's discriminative power — actual attrition rates increase sharply and monotonically across tiers.

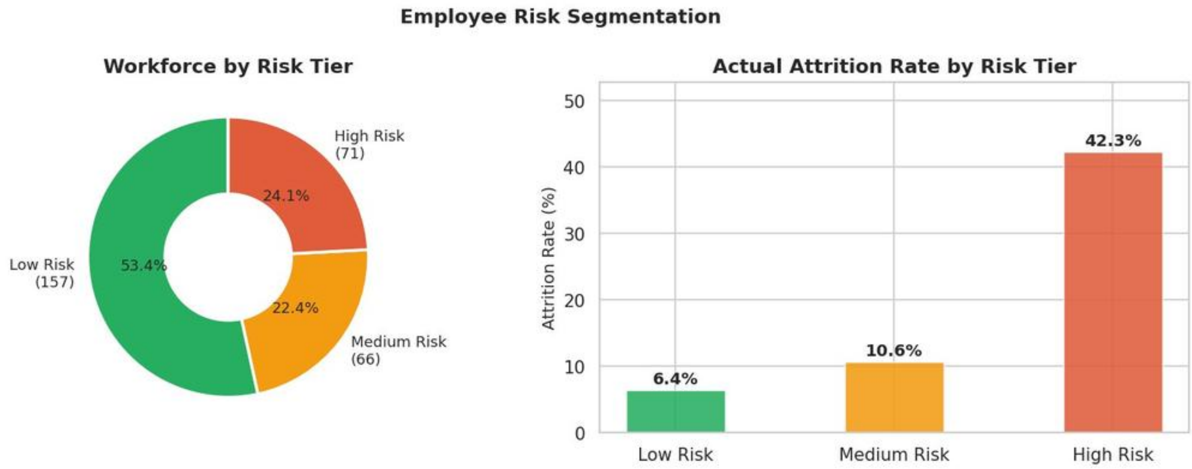

Figure 8 — Left: tier distribution. Right: actual attrition rate per tier. Low 6.4% → Medium 10.6% → High 42.3%.

Tier	Size	Actual Attrition	Avg OT Rate	Mean Income (z)	Profile
Low Risk	157 (53.4%)	6.4%	11%	+0.25	Older, long-tenured, higher income, no OT
Medium Risk	66 (22.4%)	10.6%	32%	−0.23	Mixed profile; moderate OT, below-avg income
High Risk	71 (24.1%)	42.3%	54%	−0.52	Young, short tenure, high OT, lowest income

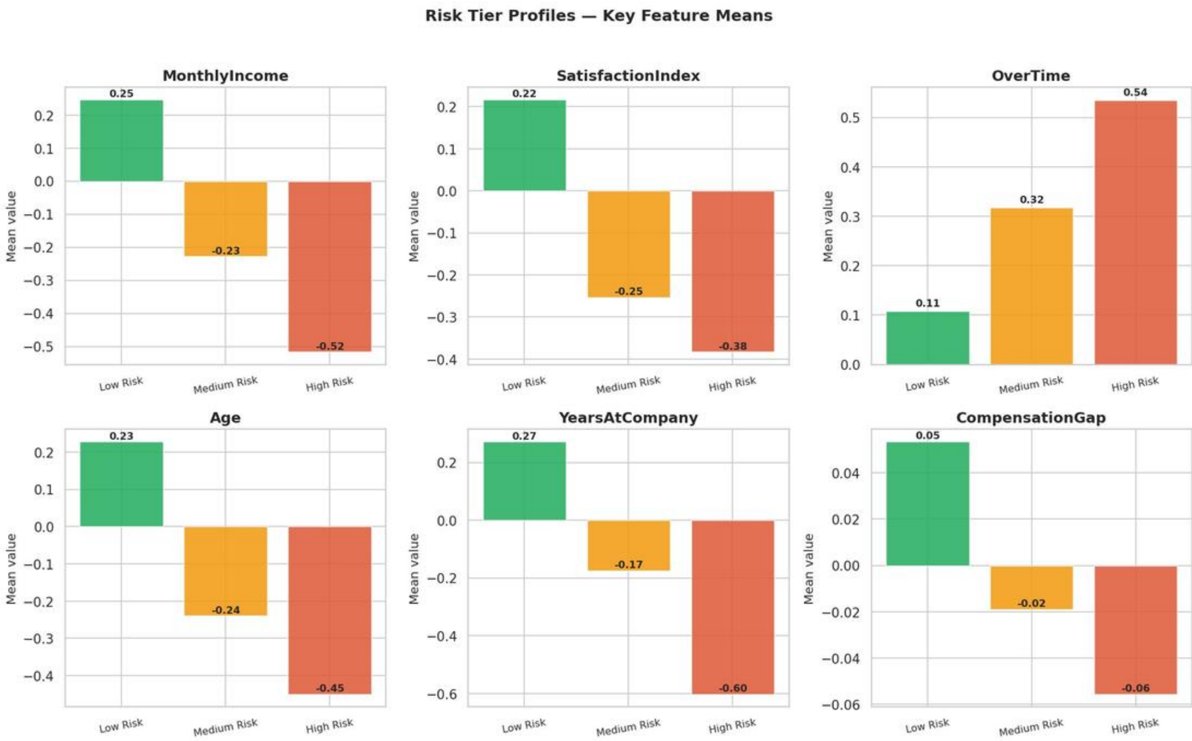

Figure 9 — Feature profiles by risk tier. All six features show clear monotonic separation across tiers.

The High-Risk tier profile is unambiguous: lowest standardised income (−0.52), lowest satisfaction (−0.38), highest overtime rate (54%), youngest and shortest-tenured employees. The six-panel profile chart confirms that every key SHAP driver separates cleanly across tiers — meaning the model's risk scores are not arbitrary but reflect genuine structural differences between groups. The Medium Risk tier is the most actionable for preventive intervention: employees here are not yet in crisis, and targeted measures (salary review, overtime monitoring) can move them to Low Risk before departure decisions are made.

6. Business Recommendations

Four recommendations follow. Each is tied to a specific SHAP-ranked feature, a specific target population defined by the segmentation, and an estimated impact.

[R1] Overtime Reduction Policy for the High Risk Tier · HIGH PRIORITY

Model Evidence	Recommended Action
OverTime is the #2 SHAP driver (0.629). The High Risk tier shows a 54% overtime rate vs 11% in Low Risk. EDA confirmed a 2.9× attrition multiplier (30.5% vs 10.4%). OverTime has the largest positive coefficient in the model (1.57 log-odds).	Cap weekly overtime at 5 hours for all employees in the High Risk tier. Require manager approval beyond this threshold. Review workload distribution in departments with highest tier-3 concentration. Pilot in one department for 90 days before full rollout.

Estimated Impact (Priority: Immediate — no new data required, policy change only): If the cap moves 30% of High Risk employees to Medium Risk, preventing 6–8 departures per year saves approximately $360K–$480K in replacement costs.

[R2] Salary Benchmarking for Negative CompensationGap Employees · HIGH PRIORITY

Model Evidence	Recommended Action
MonthlyIncome is the #4 SHAP driver (0.534). The High Risk tier shows a mean standardised income of −0.52 vs +0.25 in Low Risk. EDA confirmed a $2,002/month median gap between employees who stayed and those who left. CompensationGap also appears in top SHAP features.	Extract all employees in Medium and High Risk tiers with CompensationGap below −$500. Conduct a salary benchmarking exercise against role-level medians using existing HR pay data. Prioritise employees within their first 3 years of tenure — the peak attrition window. Present findings to compensation committee within 60 days.

Estimated Impact (Priority: Month 1–2 — all required data already exists in HR systems): Closing half the income gap for 20 flagged employees costs approximately $120K in salary adjustments. Each prevented departure saves ~$60K. A net positive outcome requires preventing just 2 departures.

[R3] Early-Career Retention Programme (Under-30 Cohort) · HIGH PRIORITY

Model Evidence	Recommended Action
TenureToAgeRatio is the #1 SHAP driver (0.635). Low ratio = young employee, short tenure = high risk. EDA showed 27.9% attrition for employees under 30 vs 10.9% for those over 40. The High Risk tier is disproportionately young (mean Age z-score −0.45) and short-tenured (−0.60).	Launch an 18-month structured mentorship and development programme targeting employees under 30 with tenure under 2 years. Include: quarterly career conversations with a senior sponsor, defined promotion criteria at 18 months, and a skills development budget. Pair with the overtime cap (R1) since this cohort overlaps heavily with the High Risk tier.

Estimated Impact (Priority: Month 3–5 — requires programme design and cohort identification): This cohort represents the highest attrition volume by count. A 20% reduction in under-30 departures prevents 15–20 per year, saving $900K–$1.2M. Programme cost (mentorship time + training budget) is estimated at $50K–$80K annually.

[R4] Satisfaction Floor Pulse Survey with Automated HR Alert · MEDIUM PRIORITY

Model Evidence	Recommended Action
SatisfactionIndex is the #17 SHAP feature (0.226) — meaningful but not dominant. However, EDA showed a clear threshold: employees scoring ≤ 2/4 on satisfaction face 19.7% attrition vs 13.9% for ≥ 3. Scores of 3 and 4 behave identically — the critical intervention point is the floor, not the ceiling. Medium Risk tier shows mean satisfaction z-score of −0.25.	Deploy a quarterly 3-question pulse survey measuring environment, job, and relationship satisfaction. Auto-flag any employee scoring a composite below 2.0. Trigger a confidential HR conversation within 2 weeks of the flag. Track manager-level aggregates to surface structural team issues rather than individual cases only.

Estimated Impact (Priority: Month 4–6 — requires tooling selection and threshold calibration): Low-cost intervention — survey tooling is negligible. Preventing 8–10 satisfaction-driven departures per year saves $480K–$600K. The signal is most reliable when combined with High or Medium Risk tier membership.

7. Implementation Roadmap

Timeline	Rec	Priority	Action
Now	**R2**	**HIGH**	Extract negative CompensationGap employees from HR database — data already exists
Month 1–2	**R1**	**HIGH**	Draft overtime cap policy; identify High Risk tier employees via IDSS model run
Month 1–2	**R2**	**HIGH**	Salary benchmarking vs. role medians; present to compensation committee
Month 2–3	**R1**	**HIGH**	Pilot overtime cap in highest-attrition department; measure 60-day attrition delta
Month 3–5	**R3**	**HIGH**	Design early-career programme; identify under-30, <2yr-tenure cohort
Month 4–6	**R4**	**MEDIUM**	Select pulse survey tooling; set 2.0 SatisfactionIndex alert threshold
Ongoing	**R5**	**MEDIUM**	Run IDSS model monthly on full employee database; distribute High Risk list to HR